Client Statistical Analysis Report

Client: Portfolio Demo / Custom CSV Project | **Prepared by:** Tianyou Liu - Statistical Analysis | **Date:** June 18, 2026

Executive Summary

High-level decision summary for business stakeholders before the full statistical methodology.

TEST	Two-Sample Z-Test for Proportions
RESULT	A statistically significant difference was detected. p = 0.0479 at alpha = 0.05.
RECOMMENDATION	Use Group_B if the business goal is to maximize conversion rate.
CONFIDENCE	95% CI for Group B - Group A: [0.0004, 0.0602]
PLAIN ENGLISH	The Two-Sample Z-Test for Proportions found a statistically significant result (p = 0.0479, alpha = 0.05). Group_B was higher than Group_A in this sample.

1. State

Define the hypotheses and significance level to evaluate the claims of the A/B test.

Significance Level (α): 0.05

Hypotheses (Symbols):

$$H_0 : p_A = p_B \quad H_a : p_A \neq p_B$$

Null Hypothesis (H_0): There is no difference in the conversion rates between Group_A and Group_B ($p_A = p_B$).
Alternative Hypothesis (H_a): There is a statistically significant difference in the conversion rates between Group_A and Group_B ($p_A \neq p_B$).

2. Plan

Identify the appropriate statistical procedure and verify its mathematical assumptions.

Recommended Procedure: Two-Sample Z-Test for Proportions

Assumption / Condition Checked	Status & Diagnostic Details
Randomness	Assuming samples are collected via independent random selection.

Independence	Samples represent less than 10% of their respective target populations.
Success-Failure Condition	Group A successes (124) & failures (876) ≥ 10? **Yes**. Group B successes (162) & failures (888) ≥ 10? **Yes**.

CONDITIONS MET

All mathematical conditions for the Two-Sample Z-Test for Proportions are satisfied. Parametric results are highly reliable.

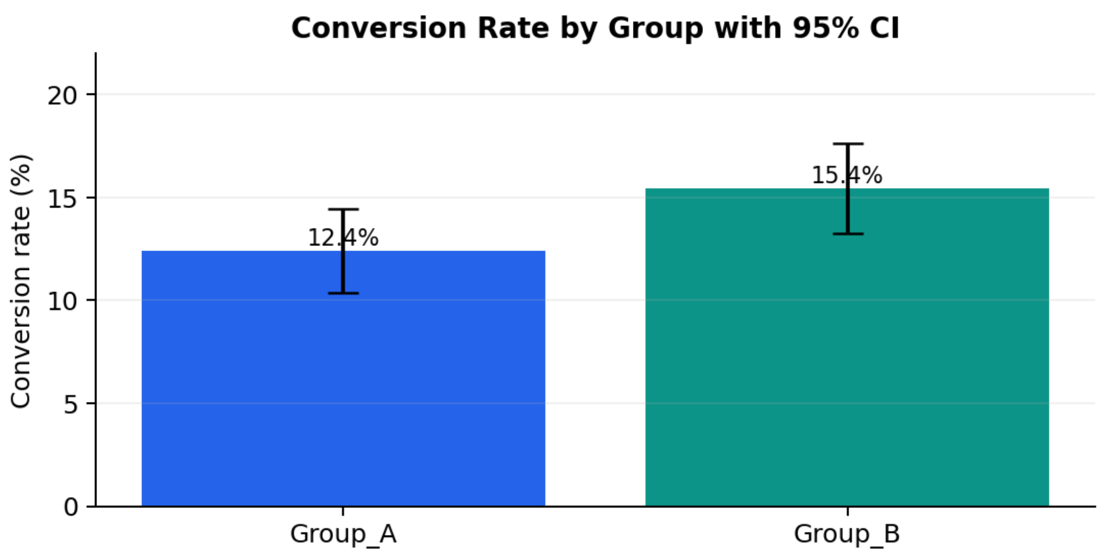
3. Do

Calculate sample metrics, standard error, test statistic, and the resulting p-value.

Sample Descriptive Statistics:

Group	Sample Size (n)	Successes (x)	Conversion Rate (p■)
Group_A	1000	124	12.4000%
Group_B	1050	162	15.4286%

Visual Summary:



Calculations & Mathematical Formulations:

• *Pooled Prop*:

$$p_{pooled} = \frac{x_A + x_B}{n_A + n_B} = 0.1395$$

• *Se*:

$$SE = \sqrt{p_{pooled}(1 - p_{pooled})(\frac{1}{n_A} + \frac{1}{n_B})} = 0.0153$$

- *Z Stat*:

$$Z = \frac{\hat{p}_B - \hat{p}_A}{SE} = 1.9782$$

- *Ci*:

$$CI = (\hat{p}_B - \hat{p}_A) \pm Z_{crit} \times SE_{diff} = (0.0004, 0.0602)$$

Statistical Parameter	Value
Test Statistic (Z)	1.97823
p-value	0.04790
95% Confidence Interval	(0.0004, 0.0602)
Risk difference	0.0303 (Odds ratio = 1.2888)

4. Conclude

State the final decision and offer a business interpretation based on the statistical results.

Compare p-value to α: 0.04790 < 0.05. Since the p-value is less than the significance level, we **reject the null hypothesis (H_0)**. There is strong evidence supporting a statistically significant difference between the two variations.

DECISION: REJECT NULL HYPOTHESIS

Business Action Recommendation: The conversion rate difference of 3.03% is statistically significant. Deploy **Group B** to production, as it is expected to generate a permanent uplift in performance.

Plain English:

The Two-Sample Z-Test for Proportions found a statistically significant result (p = 0.0479, alpha = 0.05). Group_B was higher than Group_A in this sample.